

⚖️ 7. GOVERNANCE & RELEASE GATE · STPA Hazard Analysis · Bumpless Override · Defensible Release



1. SENSING

Photons / Voltages to DMA
MIPI CSI-2 · SPI Bus Priority



2. PERCEPTION

ViT Encoders · DINOv2
3D Spatial Affordance Tokens



3. MEMORY

Temporal Belief · $SE(3)$ Trees
JEPA / RSSM World Models



5. PLANNING (System 1.5 · MPU/NPU)

Diffusion Policies · ACT Action Chunking
 H -Step Action Horizons · C^2 Jerk Continuity



4. REASONING (System 2 · MPU)

Vision-Language Foundation Models · 3D Goals
Leases with Expiring TTL (t_{expire})



6. REFLEX (System 1 · MCU)

1 kHz Real-Time Loop · Control Barrier Functions ($h(\mathbf{x}) \geq 0$) · Dynamic Stopping $d_{\text{stop}} \leq d_{\text{clearance}}$ · Veto (u_t)



THE PHYSICAL WORLD ($W_t \rightarrow W_{t+1}$) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics

↻ Closed Sensory-Motor Feedback (O_{t+1})